https://github.com/d2v6/OpenWay_Test

5.1) Describe the components of a test case. Also provide an example of a well-constructed test case.

Test Case

A **test case** is a documented set of conditions, actions, inputs, and expected results used to verify that a software application works as intended. Test cases help testers confirm that the system meets its requirements and identify defects before the software is released.

A well-written test case provides enough detail so that another tester or developer can follow the same steps, use the same data, and reproduce the same result.

Components of a Test Case

1. Test Case ID

A unique identifier assigned to each test case. This helps testers track, organize, and reference the test case easily.

Example: TC_LOGIN_001

2. Test Case Title

A short title that clearly describes what the test case verifies.

Example: Verify successful login using valid credentials.

3. Test Description

A brief explanation of what the test case is checking.

Example: Verify that a registered user can log in with a valid email address and password.

4. Priority

The importance level of the test case. Priority helps the testing team decide which test cases should be executed first, especially when time is limited.

Common priority levels:

Priority	Meaning
High	Critical functionality. Must be tested first. Failure may block major system use.
Medium	Important functionality, but not usually system-blocking.
Low	Minor functionality, UI behavior, or edge cases that have less business impact.

Example: High

5. Preconditions

The conditions that must be met before the test can be executed.

Example:
The user must already have an active registered account.

6. Test Environment

The environment, browser, device, operating system, or application version where the test is executed. This helps testers reproduce the result under the same conditions.

Example: Web application, Google Chrome browser, QA environment.

7. Test Steps

A clear, step-by-step list of actions the tester must perform.

Example:

1. Open the login page.
2. Enter a valid email address.
3. Enter a valid password.
4. Click the **Login** button.

8. Test Data

The input values required to complete the test.

Example:

Data Field	Value
Email	**testuser@example.com**
Password	**Test@123**

9. Expected Result

The result that should occur if the application works correctly.

Example:
The user should be successfully logged in and redirected to the dashboard.

10. Actual Result

The actual outcome observed during testing. This is completed after the test is executed.

Example:
The user was redirected to the dashboard.

11. Status

The final result of the test case, usually marked as **Pass** or **Fail**.

Example: Pass

12. Postconditions

The state of the system after the test case has finished.

Example: User is logged into the application and has an active session.

13. Notes

Additional information, assumptions, limitations, or reminders related to the test case.

Example: This test should also be considered for regression testing because login is a critical application function.

Well-Constructed Test Case

A well-constructed test case should be clear, complete, and easy to follow. It should include a specific objective, detailed steps, realistic test data, and measurable expected results.

A good test case should:

1. Use clear and specific language.

2. Include all necessary preconditions.
3. Include a priority level.
4. Provide steps in the correct order.
5. Make each step simple and focused.
6. Include realistic test data.
7. State the expected result clearly.
8. Be easy for another tester to repeat.
9. Be detailed enough for developers to understand the defect if the test fails.

Well-written test cases improve the quality of testing because they reduce confusion, support consistent execution, and make it easier to identify and report defects.

Example of a Well-Constructed Technical Test Case

Field	Details
Test Case ID	TC_LOGIN_001
Test Case Title	Verify successful login using valid credentials
Priority	High
Test Environment	Web application, Chrome browser, QA environment
Test Description	Verify that a registered user can successfully log in using a valid email address and password.
Preconditions	1. User account exists in the system. 2. User account status is active. 3. Login page is accessible. 4. Authentication service is available.
Test Data	Email: testuser@example.com Password: Test@123
Test Steps	1. Navigate to the application login page. 2. Verify that the email and password fields are displayed. 3. Enter testuser@example.com in the email field. 4. Enter Test@123 in the password field. 5. Click the **Login** button. 6. Observe the system response.
Expected Result	1. Login page loads successfully. 2. Email and password fields are visible

	and editable. 3. User credentials are accepted. 4. User is authenticated successfully. 5. User is redirected to the dashboard page. 6. A valid user session is created.
Actual Result	User is redirected to the dashboard page and a valid session is created.
Status	Pass
Postconditions	User is logged into the application and has an active session.
Notes	This test should also be considered for regression testing because login is a critical application function.